

Physiological and genomic insight into neuroendocrine regulation of puberty in gilts



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ARTICLE INFO

Article history:

Received 8 November 2019

Received in revised form 27 January 2020

Accepted 29 January 2020

Keywords:

Swine

Puberty

Luteinizing hormone

Kisspeptin

Genetic association

Metabolic regulation

ABSTRACT

The timing of pubertal attainment in gilts is a critical factor for pork production and is an early indicator of future reproductive potential. Puberty, defined as age at first standing estrus in the presence of a boar, is brought about by an escape from estrogen inhibition of the GnRH pulse generator, which allows for increasing LH pulses leading to the onset of cyclicity. The biological mechanisms that control the timing of these events is related to decreasing inhibitory signals with a concomitant increase in stimulatory signals within the hypothalamus. The roles of gamma-aminobutyric acid, endogenous opioid peptides, and gonadotropin-inhibitory hormone in negatively regulating gonadotropin secretion in gilts is explored. Developmental changes in stimulatory mechanisms of glutamatergic and kisspeptin neurons are important for increased LH pulsatility required for the occurrence of puberty in pigs. Age at first estrus of gilts is metabolically gated, and numerous metabolites, metabolic hormones, and appetite-regulating neurotransmitters have been implicated in the nutritional regulation of gonadotropin secretion. Leptin is an important metabolic signal linking body energy reserves with age at puberty in gilts. Leptin acting through neuropeptide Y and proopiomelanocortin neurons in the hypothalamus has important impacts on the function of the reproductive neurosecretory axis of gilts. Age at puberty in swine is heritable, and genomic analyses reveal it to be a polygenic trait. Genome-wide association studies for pubertal age in gilts have revealed several genomic regions in common with those identified for age at menarche in humans. Candidate genes have been identified that have important functions in growth and adiposity. Numerous

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Interpretive Summary: Attainment of puberty in replacement females (gilts) is critical for successful swine production, and failure of gilts to

become pubertal is a management challenge in pork production. Complex developmental changes in the hypothalamic-pituitary-ovarian (HPG) axis are necessary for puberty to occur. There is a concomitant decrease of inhibitory signals and increase of stimulatory signals in the brain that coordinates the release of reproductive hormones that trigger puberty. Puberty in the pig is metabolically gated, and metabolites and metabolic hormones such as leptin have profound effects on these regulatory mechanisms. Puberty is a heritable genetic trait, and genetic association studies have identified many genomic regions related to puberty in pigs. These genomic regions contain genes regulating growth, adiposity, and function of the HPG axis, which underscores the relationship of body composition with reproductive cycles. Some of these genes have also been associated with control of puberty in humans. Summary of the current state of knowledge about these physiological and genomic mechanisms is important to identify new solutions to manage the pubertal development of gilts in commercial swine production.

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genes regulating hypothalamic neuronal function, gonadotropes in the adenohypophysis, and ovarian follicular development have been identified and illustrate the complex maturational changes occurring in the hypothalamic–pituitary–ovarian axis during puberty in gilts.

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1. Introduction

Timing of pubertal attainment in replacement gilts is critical for successful swine production. A constant supply of cyclic gilts is necessary to meet weekly breeding needs in commercial sow farms. Timing of puberty is crucially important to ensure that replacement gilts become pubertal at optimal BW for good reproductive potential and to minimize the number of nonproductive days from selection to breeding. Age at puberty, defined in gilts as age at first standing estrus in the presence of a boar, is an important component trait of sow lifetime productivity [1–3]. Gilts that reach puberty earlier are more likely to generate a third parity and produce more pigs over their lifetime [4–6]. Gilts typically attain puberty at 6 to 7 mo of age, but 10% to 25% of gilts may fail to reach puberty, even at later ages, resulting in a number of management challenges for commercial pork production. Many factors including genetics, stress, social interactions, season, housing, and other environmental factors can impact age at puberty in gilts. These factors will not be discussed in detail here, however, as the focus of this review is neuroendocrine mechanisms regulating the activation of the hypothalamic–pituitary–gonadal (HPG) axis for initiation of puberty in gilts. Special emphasis is placed on the current state of knowledge about physiological and genomic mechanisms regulating the HPG axis during puberty. Areas where knowledge about these factors may be lacking in the pig are indicated.

Gilts can become pubertal without exposure to boars, but it is well established that exposure of gilts to boars will hasten the onset of pubertal estrus. Starting boar exposure at ages that gilts will naturally attain puberty results in a synchronous onset of puberty among gilts. Exposing gilts to boars at earlier ages (120–140 d of age) reduces the synchrony of pubertal estrus among gilts but allows for the detection of the most sexually mature gilts. Often boar exposure is confounded with factors described previously, making it difficult to isolate specific effects of boars. The studies reviewed herein use several animal models, including prepubertal gilts, sexually mature gilts, and prepubertal or sexually mature ovariectomized (OVX) gilts with or without steroid replacement. Although it may be assumed that sexually mature gilts reported in these studies have been exposed to boars, it is often not specified. The reader should consult the references in this review for a deeper understanding of the context. Nonetheless, the effects of boars on stimulating gonadotropin secretion and cyclicity in gilts [7,8] and anestrus sows [9] is expected to involve the neuroendocrine mechanisms described in this review [10].

2. Developmental changes in the HPG axis during puberty in gilts

Approximately one-third of gilts selected as breeding replacements never produce a litter [5,11,12]. Reproductive failure, including failure to cycle or maintain regular cycles, accounts for a significant number of gilt culls [13–16]. Synthesis and pulsatile secretion of the gonadotropin hormones, LH and FSH, are absolutely essential for the initiation and maintenance of reproductive cycles [17–19]. It is the synthesis and pulsatile secretion of GnRH from neurons in the hypothalamus that drives pulsatile secretion of LH, and to a lesser extent, FSH in gilts [20]. The GnRH neurons originate in the olfactory placode and migrate across the cribriform plexus to the hypothalamus. By Day 50 of gestational age, GnRH perikarya are observed in the organum vasculosum of the lamina terminalis (OVLT) in the preoptic area (POA) of the fetal forebrain of the pig [21]. By 70 d of gestational age, GnRH fibers project to the median eminence (ME) of the fetal porcine hypothalamus and are associated with capillary loops of the hypophyseal portal vasculature. In the postnatal and adult pig, GnRH neurons are localized primarily in the medial POA at the level of the OVLT, medial basal hypothalamus (MBH) including the arcuate nucleus (ARC), and in the anterior hypothalamic area (AHA) [22,23]. In the sexually mature gilt, a moderate number of GnRH neuronal cell bodies are observed in the periventricular (PeV) zone, the supraoptic nucleus (SON), and the diagonal band of Broca (dbB), with a few cells observed in the suprachiasmatic nucleus (SCN), and paraventricular nucleus (PVN).

The components of the porcine HPG axis are functional before puberty. The fetal anterior pituitary gland releases LH in response to GnRH as early as 80 to 100 d of gestational age [24,25]. Initially, serum concentrations of LH and LH pulse frequency are high until about 3 mo of age [26]. Tertiary follicles first appear on the gilt ovary between 60 and 90 d of age [27], which is when the hypothalamus and anterior pituitary gland become sensitive to estrogen negative feedback [26,28]. The negative feedback of estrogen at the hypothalamus results in the juvenile nadir, a pattern of LH secretion characterized by high-amplitude, low-frequency pulses and reduced serum concentrations of LH, which persists over the next several months until just before puberty. About 10 to 20 d before puberty occurs, the pattern of LH secretion changes to high-frequency, low-amplitude pulses [29–31] that cause maturation of ovarian follicles. This change in LH pulsatility occurs because of a reduction in sensitivity to estrogen negative feedback [32,33]. Pigs in commercial swine production that fail to reach puberty by 7 mo of age are typically administered exogenous gonadotropins (400 IU PMSG and 200 IU hCG) to stimulate growth and maturation of ovarian follicles.

Increasing serum concentrations of estrogen from pre-ovulatory Graafian follicles has a positive effect at the hypothalamus to stimulate greater secretion of GnRH that then induces an ovulatory surge of LH resulting in pubertal estrus and ovulation.

Mechanisms in the brain that bring about this age-dependent escape from estrogen negative feedback in gilts remain unclear. Expression of messenger RNA (mRNA) for *GnRH1* in the hypothalamus of gilts increases from birth to 3.5 mo of age and then is reduced 50% by 6 mo of age [34], which corresponds to the juvenile nadir in LH secretion. Between 5 and 7 mo of age, expression of *GnRH1* mRNA remained unchanged in the hypothalamus of OVX gilts given estradiol that produced clear inhibition of LH pulses at 5 mo but not 7 mo of age [35]. More recently, Thorson et al [36] confirmed that expression of *GnRH1* mRNA in the MBH of prepubertal, peripubertal, and post-pubertal gilts does not differ. This indicates that changes in *GnRH1* gene expression are not critical for the onset of puberty in gilts. Likewise, the number of cytosolic and nuclear estrogen receptors in the hypothalamus of gilts between 1 and 5.5 mo of age did not differ [37]. Expression of mRNA for estrogen receptor 2 (*ESR2*) was not different in the MBH of gilts at 3, 5, and 7 mo of age, but hypothalamic expression of estrogen receptor 1 (*ESR1*) mRNA was declined in pigs at 7 mo of age compared with 5 mo of age [35]. Hypothalamic GnRH neurons express *ESR2* [38], but not *ESR1*. Given the critical role of *ESR1* in regulating estrogen negative feedback on GnRH neurons [39], other afferent inhibitory neurons must be involved in the negative regulation of gonadotropin secretion by estrogen. The onset of puberty in gilts is thus thought to reflect an overall increase in excitatory neuronal signaling coupled with a reduction in inhibitor neuronal input to the GnRH pulse generator [40].

3. Neuronal inhibition of gonadotropin secretion

Gamma-aminobutyric acid (GABA) is the dominant inhibitory neurotransmitter in the suppression of GnRH neurons [41]. The pubertal increase in GnRH secretion in primates and rodents is proposed to be largely controlled by gene-regulated mechanisms that decrease GABAergic suppression of the GnRH neuronal system [41], and estradiol plays a biphasic role in modulating the effect of GABA on LH secretion [42]; thus, factors affecting the action of estrogen on GABAergic neurons could be important for the onset of puberty in pigs. Hypothalamic expression of mRNA for prohibitin 2 (*PHB2*), also known as repressor of estrogen receptor activity, was upregulated in 7-month-old gilts that had escaped estrogen negative feedback compared with expression in the hypothalamus of gilts at 3 and 5 mo of age that had yet to undergo this maturational change in sensitivity to estrogen [35]. Prohibitin 2 is ubiquitously expressed and is a transcriptional corepressor that acts to negatively regulate intracellular signaling of *ESR1* [43–45]. The deletion of *PHB2* alleles in mice increased the response of estrogen-sensitive tissues (uterus and mammary gland) to estradiol [46,47]. Prohibitin 2 is also a gene ontology term in the cellular ontological process of GABAergic synapses. Although GABA and GABA receptors in swine have

not been well studied in relation to the pubertal transition in GnRH/LH secretion, it is possible that the reduced suppressive effect of estradiol on LH secretion during puberty in gilts could result from *PHB2* repression of *ESR1* signaling in GABAergic neurons, which in turn reduces GABA inhibition of the GnRH pulse generator.

Endogenous opioid peptides (EOP) consist collectively of endorphins, enkephalins, and dynorphins. These EOP play important roles in suppressing LH in luteal phase gilts [48] and lactating sows [49,50]. The distribution of pro-opiomelanocortin (POMC), encoding beta-endorphin, and catecholamine neurons have been described throughout the hypothalamus of the gilt [51,52]. Secretion of LH in prepubertal OVX gilts was suppressed following intracerebroventricular (i.c.v.) treatment with morphine [53], suggesting that EOP receptors are functional and capable of affecting the GnRH pulse generator before puberty occurs. Nonetheless, there are maturational changes in the response of gilts to EOP as the general opioid receptor antagonist, naloxone, failed to increase LH secretion in prepubertal gilts [54]. Antagonizing opioid receptors with naloxone did increase LH in progesterone-treated, mature OVX gilts, but naloxone failed to affect LH secretion in prepubertal OVX gilts treated with progesterone [54]. Naloxone was effective at stimulating LH secretion in progesterone-treated prepubertal OVX gilts, however, when they reached the same chronological age as cyclic contemporary gilts. Thus, there appears to be a maturational process within the hypothalamus of gilts that allows for the development of EOP modulation of LH secretion, and this process appears to be independent of the ovary. Evidence suggests that EOP inhibition of LH secretion in the pig involves interaction with noradrenergic neurons. First, hypothalamic neurons that express POMC, tyrosine hydroxylase, and dopamine β -hydroxylase are located close together in areas that contain GnRH neurons [23,51,52]. Second, naloxone failed to increase LH secretion in sexually mature, progesterone-treated OVX gilts fed methallibure to deplete hypothalamic norepinephrine [55]. This suggests that EOP suppression of GnRH secretion in gilts involves inhibition of noradrenergic neurons. It is speculated that this is a direct effect of EOP on noradrenergic neurons, which may come about with increasing sexual maturity as has been proposed for rats and nonhuman primates [41].

3.1. Gonadotropin-inhibitory hormone

Gonadotropin-inhibitory hormone (GnIH) was identified in quail as a novel neuropeptide that suppressed gonadal development by inhibiting synthesis and secretion of LH from the anterior pituitary gland [56–58]. Neuropeptide products of the mammalian *NPVF* gene are considered homologs to avian GnIH. These neuropeptides are part of the RFamide peptide family and are characterized by the C-terminal amino acid sequence of LPXRF, where X can be L or Q [59,60] and may sometimes be referred to in the scientific literature as mammalian GnIH [61,62]. The porcine *NPVF* gene produces 3 neuropeptides designated RFamide-related peptide (RFRP)-1, RFRP-2, and RFRP-3 based on their position beginning from the N-terminus of the preprohormone [36], and the pig

hypothalamus was reported to have the greatest expression of NPVF mRNA [63]. Thorson et al [36] used an antibody to avian GnIH that recognized all 3 forms of porcine RFRP to understand the spatial distribution of these neuropeptides in the porcine hypothalamus. Positive immunoreactive staining for RFRP was observed in the posterior hypothalamus, dorsomedial hypothalamic nucleus (DMH), and ventromedial hypothalamic nucleus (VMH). Both in situ and immunocytochemical analysis demonstrate a similar pattern of RFRP expression in the VMH and DMH of rodents [64–69], sheep [70–72], and the mare [73]. The most abundant population of porcine RFRP-immunoreactive perikarya was observed in the PVN. This agrees with observations in sheep and nonhuman primates [72,74]. In sheep, RFRP fibers project from the PVN to the lateral hypothalamic area, the ARC, and the external zone of the ME [71,72]. Moderately dense RFRP fibers were observed in the lateral hypothalamic area of the pig hypothalamus, but no RFRP fibers were observed in the external zone of the ME of the gilt, which would suggest that RFRP is not released into the hypophyseal portal vasculature of the pig as has been observed in sheep [62].

Early reports indicated that RFRP-3 attenuated GnRH-induced secretion of LH from primary cultures of porcine anterior pituitary cells [75]. Furthermore, GnRH secretion from porcine hypothalamic explants was suppressed upon treatment with RFRP-3 [76]. When OVX gilts received i.c.v. infusions of RFRP-3 into the lateral ventricles of the brain, pulsatile secretion of LH remained unchanged [36], which is consistent with the effects of RFRP-3 administered into the third ventricle of OVX ewes [77,78]. Infusion of RFRP-3 into the peripheral circulation largely had no effect on secretory patterns of LH in OVX gilts unless very high doses were given [36]. It also took large pharmacological doses of RFRP-3 to affect LH secretion in intact mature boars [79]. As noted, the porcine RFRP preproprotein can yield an RFRP-2 peptide that is predicted to be 18 amino acids in length [36], and which ruminants and rodents do not produce [59,80]. Because RFRP-2 is in the same position within the preproprotein as is avian GnIH and because the amino acid sequence of porcine RFRP-2 has greater sequence homology to avian GnIH than RFRP-3, it was speculated that RFRP-2 was a porcine-specific GnIH. However, when RFRP-2 was infused into the peripheral circulation of OVX gilts, LH pulses were unaffected [36].

The NPVF-derived RFRP neuropeptides bind and activate neuropeptide FF receptor (NPFFR)-1 and NPFFR-2 [81–83]. The expression of NPFFR1 in the MBH did not differ between prepubertal, peripubertal, and postpubertal gilts, but hypothalamic expression of NPFFR2 was increased with the onset of puberty [36], although the biological consequence of this is presently unclear. In the anterior pituitary gland, there was a dramatic decrease in expression of NPFFR1 and NPFFR2 mRNA with the onset of sexual maturity [36], which agrees with observations in chickens and rodents [84,85]. It is noted that the effects of RFRP on LH secretion in mammals has been highly inconsistent. Several reports indicate that RFRP-3 can inhibit [68,72,86,87], have no effect [64,73,77,78,88], or even stimulate LH [89]. Sometimes, opposite effects of RFRP-3 on LH have been observed in the same experimental models, which has led

some authors to suggest that mammalian RFRP are not GnIH peptides [64,78]. The strong aforementioned in vitro data that support a role of RFRP in suppressing LH and GnRH secretion in the pig and the reduced expression of NPFF receptors in the anterior pituitary gland of the gilt during puberty would suggest that reduced inhibition of RFRP on LH secretion from gonadotropes could contribute to increased LH secretion. Nonetheless, RFRP does not appear to have a potent suppressive effect on in vivo secretion of LH in pigs, either centrally or peripherally, and thus does not appear to act as a hypophysiotropic GnIH in pigs. It therefore remains unclear what role RFRP plays in puberty onset in gilts as has been hypothesized for nonhuman primates [41].

4. Neuronal stimulation of gonadotropin secretion

The catecholamines dopamine, epinephrine, and norepinephrine are biogenic monoamines that function in the brain as neurotransmitters. Tyrosine hydroxylase is the rate-limiting step in the synthesis of dopamine, which is subsequently converted to epinephrine and norepinephrine by dopamine β -hydroxylase. Neurons containing tyrosine hydroxylase are localized within the PeN, SON, PVN, and ARC of the pig hypothalamus [51]. Immunostaining for dopamine β -hydroxylase in the pig hypothalamus was observed as fibers and punctate varicosities throughout these same areas. Catecholamines are generally considered to be stimulatory for LH secretion, but their role in regulating LH secretion in pigs is complex. Bromocriptine is a potent dopamine receptor agonist and secretion of LH increased, as expected, in sows treated with bromocriptine during late gestation and lactation [90,91]; however, bromocriptine treatment suppressed LH during mid-gestation [92] and had no effect on the secretion of LH in sexually mature OVX gilts [93]. Injection of norepinephrine in the lateral ventricles of the brain stimulated a short-duration pulse of LH in adult gonadectomized Göttingen miniature pigs [94], but norepinephrine had both stimulatory and inhibitory effects on LH, dependent on what area of the porcine hypothalamus received norepinephrine microinjection [95]. Terasawa and Fernandez [41] reviewed supporting evidence of norepinephrine in the onset of puberty in laboratory animals, which included evidence that catecholamine synthesis blockers delay the timing of puberty in rats and that norepinephrine in the POA and ME in laboratory animals increases around the time of puberty. Methallibure is an inhibitor of dopamine β -hydroxylase and has been used to study the effects of hypothalamic depletion of norepinephrine on the secretion of LH [96]. The pulsatile secretion of LH in sexually mature OVX gilts was completely abolished when they were fed methallibure [97]. Furthermore, the estradiol-induced surge of LH was completely blocked in methallibure-treated gilts, and this effect was observed regardless of whether supplemental GnRH was given to maintain basal LH pulses and pituitary secretory response to GnRH [97]. These data suggest that both pulsatile and surge secretion of LH in the pig is dependent on norepinephrine-stimulated secretion of GnRH.

Glutamate and aspartate are excitatory neurotransmitters that are important for GnRH and gonadotropin secretion. These excitatory amino acids stimulate GnRH release by acting on the glutamatergic N-methyl-D-aspartate (NMDA) receptor. Activation of NMDA receptors with N-methyl-D L-aspartate (NMA) accelerates puberty in laboratory rodents and nonhuman primates [41,98], and the role of NMA in the secretion of gonadotropin hormones in pigs has been previously reviewed [99]. Consistent with the importance of NMDA receptor activation for puberty, NMA increased LH concentrations in prepubertal gilts [100] and both LH and FSH in lactating sows [101]. This stimulatory effect of NMA is dependent on GnRH [101], indicating a hypothalamic site of action. In contrast, NMA decreased LH secretion in sexually mature OVX gilts [102–104], indicating the effect of NMA in gilts is dependent on the hormonal milieu of ovarian steroids. There was no effect of NMA on serum concentrations of LH in OVX gilts treated with estradiol [102] or in gilts during the follicular phase [104]. When OVX gilts were primed with luteal phase concentrations of progesterone, NMA treatment reduced LH concentrations [102], but NMA failed to suppress mean concentrations of LH in luteal phase gilts [104], which also have low levels of estrogen that may have altered the LH secretory response. These data suggest therefore that increasing glutamatergic tone is important for increased LH secretion during escape from estrogen negative feedback in gilts. In this regard, 90% of kisspeptin neurons in the ARC are also glutamatergic [105].

4.1. Kisspeptin is essential for reproduction in pigs

Kisspeptin was found to be important for normal reproduction when it was discovered that mutations in the gene for the kisspeptin receptor resulted in a condition of hypogonadotropic hypogonadism, which is characterized by a lack of gonadal development and very low levels of gonadotropin secretion in patients that fail to go through puberty [106–108]. Boars in which a functional kisspeptin receptor was knocked out with gene-editing technology recapitulated this nonpubertal hypogonadal phenotype [109]. Seminiferous tubules formed in the testes of these kisspeptin receptor knockout boars, but no spermatozoa were observed in the lumen of the tubules. Kisspeptin receptor knockout boars had small testes with very low circulating concentrations of testosterone, which was presumed a consequence of gonadotropin insufficiency. When these boars were treated with GnRH or gonadotropin hormones (LH and FSH), the testicular size was increased more than 200%, but sperm production was not restored. These data strongly support the expectation that kisspeptin signaling in pigs is absolutely required for puberty.

The localization of kisspeptin expression in the hypothalamus of the pig has not been fully characterized [110]. The *KISS1* gene is expressed in the MBH and PeV of gilts [36,111]. Within the ARC of the pig, a distinct spatial distribution of *KISS1* is seen, with the greatest expression in the mediocaudal sections [111,112]. Recent immunocytochemistry data illustrate that neuronal cell bodies and nerve fibers for kisspeptin are evident in the porcine ARC [113]. Kisspeptin neurons in the ARC of rodents and sheep

coexpress neurokinin B (NKB) [114,115], also known as tachykinin 3 (TAC3), that acting through its cognate receptor (TAC3R) is thought to be an important regulator of LH secretion in prepubertal ewe lambs [116]. Similarly, virtually all kisspeptin neurons in the ARC of the pig coexpress NKB [113]. Given the similarity in kisspeptin and NKB expression in the porcine ARC with that of sheep, it is likely that NKB affects LH secretion in prepubertal gilts as well, but this remains to be tested.

The role of NKB and its receptor in regulating puberty of the pig is not presently understood. In this regard, expression of TAC3R has recently been localized throughout several areas of the hypothalamus in OVX gilts [113]. Roles of TAC3R in various areas of the hypothalamus are only partially elucidated in most species; however, many of the hypothalamic regions that display TAC3R immunostaining in the pig also contain GnRH neurons. The dbB and the PVN had the greatest number of TAC3R expressing cells, followed by the POA and SCN. Within the PeV and RCh, TAC3R immunostaining was also evident. Administering a TAC3R agonist (senktide) into the POA or the RCh of sheep elicits a surge-like release of LH [117]. Expression of TAC3R containing neurons within the PeV of gilts is intriguing, given that both GnRH and kisspeptin neurons are located in this area of the pig hypothalamus [23,111]; however, the identification of the TAC3R containing neurons in the porcine PeV has not yet been determined. In mice, TAC3R has been shown to be expressed in GnRH neurons [39,118], but GnRH and TAC3R immunostaining did not colocalize in the pig hypothalamus [113], which agrees with data in the ewe [119]. About 40% of GnRH neurons were in close apposition to TAC3R containing cells, but close contacts were few, suggesting that the TAC3R containing cells in the porcine hypothalamus are not directly regulating GnRH neurons in pigs. The ARC of gilts was notable for the absence of TAC3R immunostaining [113]. This stands in sharp contrast to the clear expression of TAC3R in the ovine ARC [119]. The functional relevance to this species-specific difference in TAC3R in relation to puberty is not yet evident.

Kisspeptin has potent stimulatory action on gonadotropin secretion in gilts. A robust secretion of LH and FSH followed i.c.v. treatment of prepubertal gilts with kisspeptin [120]. When kisspeptin was injected into the peripheral circulation of prepubertal gilts, a surge-like secretion of LH but not FSH was observed [120,121]. This surge of LH lasted less than 1.5 h and reflects the short 60 s half-life of kisspeptin. The lowest dose of kisspeptin (11.4 nmol/kg, 768 nmol per gilt) is considered maximal as higher doses (3,839 and 7,678 nmol per gilt) did not produce any greater or more sustained release of LH. In sheep and cattle, doses ranging from 0.75 to 5 nmol/kg are sufficient to produce a similar surge-like release of LH [122–124] to that observed in prepubertal gilts. Because of the short half-life, repeated hourly injections of kisspeptin were required to stimulate a surge release of LH (17 h after the start of treatment) to induce ovulation in prepubertal ewe lambs [124]. Recently, a potent kisspeptin receptor agonist (C6) that has a much longer half-life has been developed [125]. A dramatic surge-like increase in LH lasting for more than 16 h was induced in 18-week-old

prepubertal gilts that had been given progesterone (altrenogest) and gonadotropins (PMSG and hCG) to induce follicular growth before C6 treatment [126]. Sustained release of LH after C6 treatment caused ovulation in 5 of the 7 treated gilts. There are no data, that we are aware of, as to whether sensitivity to kisspeptin changes with sexual maturation in gilts. Expression of the kisspeptin receptor gene increases at puberty in rats [127], and this increased expression of kisspeptin receptor is thought to be partly responsible for increasing LH pulsatility, leading to puberty in primates [128]. Expression of mRNA for kisspeptin receptor in the porcine hypothalamus differs with the stage of the estrous cycle, indicating regulation by ovarian steroids [129]. It therefore remains a distinct possibility that increased sensitivity to kisspeptin is a key part of pubertal maturation in the pig.

4.2. Developmental changes in kisspeptin

Redmond and coworkers [130] showed that as ewe lambs reached pubertal ages, increased expression of the *KISS1* gene in the ARC was positively related to the frequency of LH pulses. In gilts, expression of *KISS1* in the ARC during pubertal development is robust but does not change with the stage of sexual maturation [36,131]. Moreover, expression of *KISS1* in the ARC did not differ between sexually mature OVX gilts, given estradiol or OVX control gilts [111]. Thus, the regulation of ARC expression of *KISS1* during puberty in the gilt is different from the ewe lamb and appears to be independent of estrogen feedback. It should be noted, however, that expression of *TAC3* and *TAC3R* genes is upregulated in the MBH of peripubertal gilts compared with prepubertal gilts [36]. It is speculated that reduced sensitivity to estrogen negative feedback involves increased neurokinin B activation of kisspeptin neurons, rather than a change in *KISS1* expression, for the upregulation of LH pulses and puberty in gilts. Thus, it is still expected that kisspeptin neurons in the ARC play a critical role in pubertal onset in pigs.

Estradiol has a biphasic effect in gilts, inhibiting basal LH pulses through negative feedback before stimulating an ovulatory surge of LH via positive feedback [132,133]. In mice, the positive feedback effects of estradiol on LH secretion involve kisspeptin neurons in the anteroventral periventricular nucleus (AVPV). Livestock do not have an AVPV, rather kisspeptin neurons are dispersed throughout the POA. When sexually mature OVX gilts were given a dose of estradiol sufficient to stimulate an ovulatory surge of LH, expression of *KISS1* in the PeV was upregulated compared with control OVX gilts [111]. It is speculated that separate populations of kisspeptin neurons in the ARC and the PeV of the gilt mediate negative and positive estrogen feedback for the control of tonic and surge secretion of LH, respectively. In this regard, kisspeptin neurons in the PeV may be an important component of the LH surge center in the pig [133]. During development, between 0 and 5 mo of age, *KISS1* expression in the PeV of gilts was undetectable [131]. It is unknown if expression of *KISS1* in the PeV of gilts is upregulated at later ages as gilts become pubertal.

5. Nutritional regulation of gonadotropin secretion in gilts

It is well recognized that the onset of puberty in livestock species is metabolically gated. Gilts with greater growth rates reach puberty at earlier ages [134]. Similarly, restricting feed or limiting dietary energy and lysine results in gilts having delayed onset of puberty [135–137]. Modern maternal-line gilts have greater growth rates and larger mature BWs than in previous generations, yet they are much leaner. Overall average age at first estrus of gilts in commercial production has not changed all that much because of industry protocols that specify boar exposure and estrus detection not commence until 170 to 180 d of age. Nonetheless, increasing leanness of replacement gilts has negative effects on reproduction, including expression of estrus [138]. Moreover, reduced body fat reserves may limit the ability of gilts to maintain reproductive cycles in breeding when diets are changed, and growth slows [139]. This is typically manifest in gilts exhibiting reproductive cycles but subsequently becoming acyclic, which results in producers inducing a return to cyclicity with exogenous gonadotropins.

Unlike laboratory rodent species, short-term fasting does not affect pulsatile secretion of LH in gilts [140,141]; however, feed restriction of 7 to 10 d in duration is sufficient to reduce the frequency and increase the amplitude of LH pulses in sexually mature and prepubertal OVX gilts [112,142]. Thorson et al [112] showed that this nutrition-induced reduction in LH pulse frequency was not related to changes in expression of *KISS1* in the ARC. Prolonged nutrient restriction (100 d) of sexually mature gilts, however, abolishes LH pulses leading to acyclicity [18] and decreases expression of *KISS1* and its receptor in the MBH [143]. On the other hand, feeding gilts diets with excess metabolizable energy to promote early attainment of puberty increased expression of *GnRH1*, *KISS1*, and kisspeptin receptor in the POA, but not in the MBH [143]. Similar to observations for estradiol, separate populations of kisspeptin neurons in the hypothalamus of gilts may mediate the effects of nutrition on the reproductive neurosecretory axis. It does not appear that the negative effects of undernutrition on secretion of LH in the gilt involve transcriptional changes in *KISS1* within the ARC.

Given that the hypothalamus senses metabolic state for the proper activation of gonadotropin secretion, a great deal of effort has been placed on identifying and understanding what signals may be mediating these effects. A host of metabolites, metabolic hormones, and neurotransmitters affecting appetite and feed intake have been linked to nutritional regulation of LH secretion in gilts (Table 1). Blood glucose concentrations, which reflect energy metabolism, are probably most critical in this regard as inhibiting glycolysis in gilts suppresses LH secretion [141]. Similarly, the effects of insulin, free fatty acids, and other metabolites are a consequence of their influence on plasma concentrations of glucose. Altering energy balance of gilts results in repartitioning of nutrients to maintain euglycemia, causing dramatic changes in serum concentrations of these metabolic factors without affecting the expression of *KISS1* in the ARC [112]. Circulating concentrations of IGF-I

Table 1

Putative metabolic and neuroregulatory factors associated with secretion of gonadotropin-releasing hormone (GnRH) and luteinizing hormone (LH) from the hypothalamus and anterior pituitary gland, respectively, in gilts.

Regulator	Animal model	Experimental model	Dose(s)	Hormone parameter	Effect	Reference
Leptin	Prepubertal gilt	ICV	10, 50, 100 µg	LH	No effect	[144]
	Prepubertal gilt	cultured pituitary cells	10 ⁻¹⁴ to 10 ⁻⁶ M	LH	Increased	[144]
	Prepubertal gilt	Hypothalamic explant	10 ⁻¹² , 10 ⁻¹⁰ , 10 ⁻⁶ M	GnRH	Increased	[144]
NPY	Prepubertal gilt, OVX	ICV	10, 50, 100 µg	LH	Decreased	[145]
	Prepubertal gilt	Cultured pituitary cells	10 ⁻⁹ to 10 ⁻⁶ M	LH	No effect	[146]
MSH	Prepubertal gilt, OVX	SHU9119, ICV	0.01, 0.1, 1 nmol	LH	No effect	[147]
	Prepubertal gilt, OVX	NDP-MSH, ICV	0.03, 0.3, 3 nmol	LH	No effect	[147]
			10 µg	LH	Decrease	
Nesfatin-1	Prepubertal gilt	ICV	100 µg	LH	Increased	[148]
Orexin-B	Prepubertal gilt	Cultured pituitary cells	1, 10, 100 nM	LH	Increased	[149]
Urocortin	Prepubertal gilt, OVX	ICV	5, 50 µg	LH	Decreased	[150]
IGF-I	Prepubertal gilt	ICV	25, 75 µg	LH	No effect	[151]
	Prepubertal gilt, OVX	ICV	0.1, 1, 10, 25, 75 µg	LH	No effect	[152]
Insulin	Mature gilt	Cultured pituitary cells	0.01, 1 nM	LH	Increased	[153]
	Prepubertal gilt, OVX	ICV	1, 10, 100 µg	LH	No effect	[152]
	Mature gilt, OVX	ICV	6 ng	LH	Increased	[152]
	Prepubertal gilt	Cultured pituitary cells	10 ⁻¹¹ to 10 ⁻⁸ M	LH	Decreased	[152]
Glucose	Prepubertal gilt, OVX	2DG, IV	100, 300 mg/kg BW	LH pulses	Decreased frequency	[141]
	Prepubertal gilt	IV, GnRH challenge	1 g/kg BW	LH surge	Decreased GnRH stimulation	[154]
	Prepubertal gilt, OVX, Diabetic gilt, OVX	IV, nutrient restriction No insulin	622.5 g	LH pulses LH surge LH pulses	Increased Abolished estradiol induction No effect	[155] [156]
	Diabetic gilt	No insulin		LH pulses	Increased	[157]
FFA	Prepubertal gilt	Cultured pituitary cells	300–600 mg/dL	LH	GnRH stimulation decreased	[152]
	Prepubertal gilt	10% Lyposyn II, IV	3 mL/kg BW; hourly for 9 h	LH pulses	Increased amplitude	[154]
	Prepubertal gilt	Cultured pituitary cells	Linoleic acid 10 ⁻⁶ to 10 ⁻⁴ M	LH	Increased basal release, GnRH stimulation decreased	[152]
			Oleic acid 10 ⁻⁹ , 10 ⁻⁷ , 10 ⁻⁵ M			

Abbreviations: 2DG, 2-deoxy-D-glucose, a competitive inhibitor of glycolysis; FFA, free fatty acid; ICV, intracerebroventricular injection; IV, intravenous injection; MSH, melanocyte-stimulating hormone; NDP-MSH, melanocortin 3/4 receptor agonist; OVX, ovariectomized; SHU9119, melanocortin 3/4 receptor antagonist.

Lyposyn II (10%) = lipid emulsion containing linoleic (65.87%), oleic (17.7%), palmitic (8.8%), linolenic (4.2%), and stearic acids (3.43%).

increase during pubertal maturation in the pig [158]. The positive reproductive effects of IGF-I in gilts result from increased sensitivity of the pituitary gland to GnRH [144] and local ovarian effects promoting follicular steroidogenesis [159]. This suggests that the effects of these factors on the gonadotropic axis of developing gilts is permissive rather than directly regulatory.

5.1. Leptin regulation of gonadotropin secretion

Perhaps no metabolic hormone is more important for reproduction than leptin [160,161]. Circulating concentrations of leptin increase with age in gilts and are positively associated with the amount of backfat [162]. Both adipose expression of the leptin gene and circulating concentrations of leptin are reflective of energy balance in pigs [112,142,163]. During escape from estrogen negative feedback, expression of leptin mRNA in adipose tissue of gilts increases approximately 4-fold, which is driven by serum concentrations of estradiol but not BW [164]. Leptin receptor is expressed in the pig pituitary gland and hypothalamus at the level of the ARC [165,166]. Hypothalamic neurons in the POA, PVN, SOP, LHA, DMH, and VMH of the pig also express leptin receptor [167,168]. The identification

of most of these leptin receptor-expressing neurons are not known, but their organization in the porcine hypothalamus aligns with GnRH and kisspeptin expression, suggesting that they may be involved with gonadotropin secretion in gilts. Indeed, leptin stimulates the release of GnRH from hypothalamic explants of pigs [144], and serum concentrations of leptin have a positive genetic association with age at puberty in gilts [169]. Neurons for GnRH, however, lack leptin receptor [170,171], suggesting leptin potentially uses kisspeptin to stimulate LH secretion. The ob/ob mouse, which lacks endogenous leptin, is hypogonadotropic and has reduced expression of *KISS1* in the ARC compared with wild-type mice [172,173]. Leptin receptor mRNA is expressed in kisspeptin neurons within the ARC of guinea pigs and mice [172–174]. In addition, kisspeptin neurons within the ARC colocalized with STAT3, the major intracellular signal through which leptin receptor action is mediated, and leptin stimulated firing of kisspeptin neurons in ARC slices of guinea pigs [172,174]. Just as negative energy balance seems to affect kisspeptin cells in the ARC but not the AVPV, so too does leptin. Louis et al [171] reported that few kisspeptin cells in the AVPV expressed leptin receptor. Furthermore, kisspeptin neurons in the AVPV do not colocalize with STAT3 [172], and leptin fails to

stimulate increased *KISS1* expression in the AVPV [105,172,173]. Thus, it seems clear that leptin acts within different subpopulations of neurons to control LH secretion. Increasing concentrations of leptin during pubertal development of gilts could stimulate kisspeptin neurons in the ARC indirectly through neuropeptide Y (NPY) and POMC neurons to increase the activity of the GnRH pulse generator for puberty to occur [175].

5.2. Neuropeptide Y and melanocortin

Neuropeptide Y acts centrally to suppress LH secretion in gilts [145], implying that leptin-induced reductions in NPY could be related to increased pulses of LH at puberty. In ruminants, between 30% and 60% of kisspeptin neurons in the ARC are in close apposition to NPY fibers [176]. In heifers, NPY fibers also contact GnRH neurons in the POA and MBH [177]. Feeding heifers high energy diets to promote early puberty reduced the number of NPY fibers contacting GnRH but not kisspeptin neurons [177]. Before puberty, it is expected that NPY suppresses LH pulse frequency by inhibiting both GnRH and kisspeptin cells in the gilt hypothalamus. Increasing concentration of leptin during development, in turn, reduces NPY inhibition of GnRH neurons to allow for the increasing frequency of GnRH release necessary to achieve puberty.

Given the importance of EOP in regulating LH secretion in gilts, and the role of melanocortin receptors 3 and 4 (MC3/4R) for regulating energy balance and hypothalamic gene expression in the pig [147,178], the potential of POMC interaction with GnRH or kisspeptin neurons is particularly interesting. In gilts, POMC perikarya are localized specifically within and around the immediate location of the ARC [52]. Fibers for POMC were noted as projecting rostrally from the ARC to the SCN, SOP, LHA, medial POA, and dbB. Gonadotropin-releasing hormone neurons in the hypothalamus of gilts showed numerous close contacts with POMC-containing varicosities, and both GnRH and POMC fibers have extensive overlap within the ME [179]. Neuronal fibers containing alpha-melanocyte-stimulating hormone (α MSH), a product of the *POMC* gene, contacted 20% of GnRH neurons in the POA and MBH in prepubertal heifers [180]. Similarly, 40% to 60% of kisspeptin neurons in the heifer ARC were contacted by α MSH containing fibers. Feeding heifers high-energy diets to increase circulating concentrations of leptin did not affect the proportion of GnRH neurons contacted by α MSH but increased the number of kisspeptin neurons in the ARC with α MSH contacts. Furthermore, about 25% of kisspeptin neurons in the mouse coexpress MC4R [105]. These data suggest that leptin stimulation of POMC neurons could indirectly upregulate kisspeptin signaling for increased LH pulses near puberty. Consistent with this, i.c.v. infusion of a melanocortin agonist (MTII) increased LH secretion and *KISS1* expression in the POA of luteal phase ewes [181]. In addition, SHU9119, an MC3/4R antagonist, attenuated leptin potentiation of estradiol-induced LH surges in rats [182]. In contrast, neither i.c.v. treatment of prepubertal OVX gilts with a melanocortin agonist (NDP-MSH) or SHU9119 affected LH secretion [147]. It is noted that MTII does not restore fertility in ob/ob mice nor does SHU9119 block

leptin stimulation of LH secretion in the leptin-deficient mouse [183]. The hypothalamic organization of POMC neurons and fibers within the porcine hypothalamus supports the expectation that leptin stimulating α MSH action at GnRH and(or) kisspeptin neurons affect LH secretion, but further research is required to substantiate this hypothesis.

6. Genomic insights into mechanisms of puberty

Age at puberty is one of the few reproductive traits that is moderately heritable, with a mean estimate in pigs of 0.33 [184]. Using age at puberty as a selection criterion, realized heritability was estimated at 0.25, and age at first estrus was reduced by 16 to 20 d [185,186]. Genomic studies rely on well-pedigreed resource populations in which large numbers of animals (typically >1,000) have been carefully phenotyped for age at first estrus. Numerous quantitative trait loci (QTL) were identified in studies that sought to understand the genomics of puberty as a quantitative trait [187–190]. Lack of sufficient genetic markers were limitations of these early studies, and the identified QTL were large, which constrained their usefulness for identifying specific regulatory factors. Nonetheless, these studies illustrated that puberty in pigs was a complex polygenic trait, being controlled by many genes that each have small effects.

Targeted genotyping has been used to interrogate relationships of specific genes with age at puberty. Several polymorphisms in the kisspeptin receptor gene were identified in a White Duroc $\times$ Chinese Erhualian cross-bred resource population [129]. Five haplotypes were examined for genetic association with age at puberty. Despite significant variation in age at first estrus in this population, no association was detected. Given the key function of kisspeptin in regulating gonadotropin secretion and the critical importance of kisspeptin receptors for the attainment of puberty, this result may seem counterintuitive. Genetic variants in genes such as kisspeptin or GnRH, which are essential for reproduction, often occur in livestock at low allele frequencies that limit the identification of associations with phenotypic traits. Neuropeptides are typically small with short coding sequences that limit the detectable variation. In the case of kisspeptin, rare sequence variants are often heterozygous, which allows for the normal allele to compensate for the loss of function [191]. Perhaps more importantly, functional variants in kisspeptin or its receptor that have deleterious effects are not likely to increase in allele frequency because animals harboring such mutations would be culled from the breeding herd for infertility. In a more recent study, potential functional variants in the *NPFFR2* gene were identified in a Yorkshire $\times$ Landrace population [36]. Three SNP predicted to be damaging to receptor function were associated with differences in age at first estrus. As previously discussed, physiological evidence that NPFF receptor ligands are major regulators of gonadotropin secretion in gilts is not very convincing. This fact may be reflected in the relatively small allele substitution effect of 1.5 to 2 d for the nonsynonymous SNP in *NPFFR2*. These examples illustrate that genomic associations, or the lack thereof, with specific genes cannot simply be equated

Sexual Maturation in the Gilt

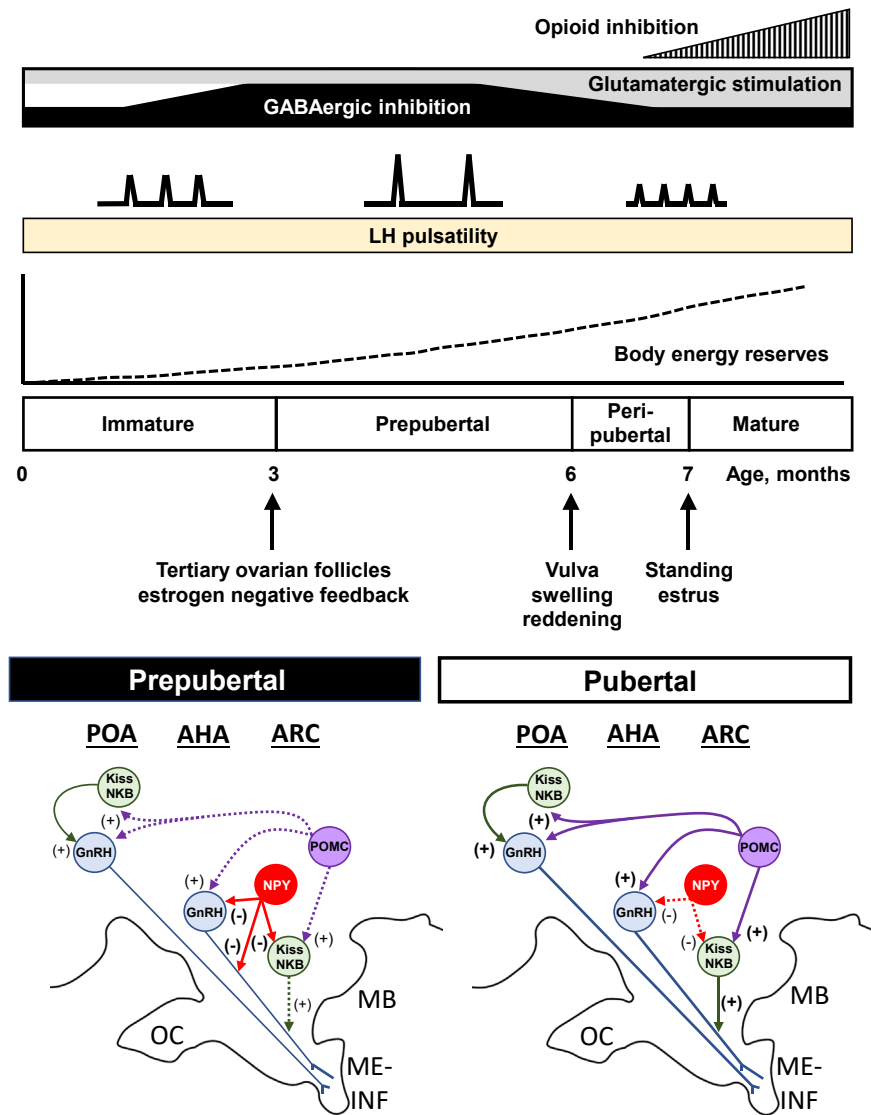


Fig. 1. Schematic illustration of growth, metabolic and neuroendocrine changes in the hypothalamus for the occurrence of puberty in gilts. Early postnatally, circulating concentrations of LH are high as a result of GnRH causing frequent pulses of LH. Increasing GABAergic inhibitory tone and the development of hypothalamic sensitivity to estrogen negative feedback is manifest in LH pulses with high amplitude but low frequency during the juvenile prepubertal period. As gilts grow, they gain increasing body energy stores, primarily in the form of adipose tissue, that must reach a critical threshold by an appropriate age for puberty to occur. A reduction in GABA inhibition and a concomitant increase in glutamatergic stimulation of the GnRH pulse generator for high-frequency, low-amplitude pulses of LH during the peripubertal period precede the spontaneous manifestation of pubertal estrus. Sufficiency of body energy reserves is signaled to the hypothalamus by neuropeptide Y and pro-opiomelanocortin cells in the arcuate nucleus (ARC), which act as metabolic sensors for activation of GnRH secretion. Increasing leptin inhibits the negative effects of NPY on GnRH and kisspeptin/neurokinin B (Kiss/NKB) neurons and upregulates activation of POMC cells, which project axonal fibers rostrally to the preoptic area and anterior hypothalamic area (AHA). These afferent projections are predicted to impact both Kiss and GnRH expression cells in the POA and ARC, thus driving increased secretion of GnRH for high-frequency, low-amplitude pulses that promote ovarian follicular growth and maturation. Development of opioid inhibition of LH secretion in gilts occurs after pubertal maturation. OC, optic chiasm; MB, mammillary body; ME, median eminence; INF, infundibulum.

to physiological importance in the regulation of gonadotropin secretion for puberty. Thus, it has been challenging to use targeted genotyping to gain much insight into mechanisms controlling activation of the HPG axis for puberty in gilts.

6.1. Genome-wide association studies and candidate genes

Sequencing technologies have rapidly evolved over the last decade and offer new opportunity to identify potential physiological mechanisms regulating puberty that may

have before been unrecognized. The porcine beadchip array, containing 60,000 genetic markers equally dispersed across the swine genome, has been used in genome-wide association studies (GWAS) to identify new QTL for age at first estrus in gilts [192–194]. Marker heritability estimates range from 0.32 to 0.38 and explain up to 26% of the phenotypic variance in age at puberty [192,194]. Numerous candidate genes were identified in these QTL associations, and 6 QTL associated with age at first estrus in pigs contained candidate genes associated with age at menarche in humans [195,196].

Many genes associated with age at puberty in both humans and pigs are related to growth, adiposity, and reproduction (eg, *CRTC1*, *IQCH*, *RORA*, *AQP8*, and *NEGR1*). The *CRTC1* gene is a cAMP-response element binding (CREB) protein-regulated transcription coactivator, that can act in the ovary to regulate CREB-activated genes such as steroidogenic acute regulatory protein, which is necessary for steroidogenesis [197]. Leptin activates *CRTC1* for recruitment to the promoter of CREB-activated genes. Knocking out *CRTC1* in mice caused reproductive failure resulting from severely reduced *KISS1* expression in the hypothalamus [198]. Although it was subsequently reported that *CRTC1* was not essential for fertility [199], *CRTC1* deletion caused dramatic effects on metabolic homeostasis that could negatively affect the expression of kisspeptin. The *RORA* gene is a retinoic acid-related orphan receptor that is involved in cerebellar development, adipogenesis, lipid and steroid metabolism, and regulation of clock genes. A naturally occurring loss of function mutation of *RORA* in mice leads to delayed puberty, irregular cyclicity, and shortened reproductive life [200]. Finally, neuronal growth regulator 1 (*NEGR1*) is highly expressed in the hypothalamus where it promotes neurite growth and synapse formation. It has been linked through GWAS to obesity and body mass index in humans [201–203], and deleting *NEGR1* in mice reduced BW by decreasing food intake [204]. Other genes, including *MTCH2*, *ZNF608*, *ADAMTSL3*, and *PRDK1*, associated with pubertal age in gilts [194], were also associated with body mass index, obesity, and height in humans [202,205–208], illustrating the interrelationship of growth, body composition, and puberty in pigs at the genomic level.

Tart et al [192] uncovered *AVPR1A* as potentially important for pubertal estrus in gilts. Arginine vasopressin receptor 1A is in the same family as oxytocin receptors, and many biological functions of oxytocin are mediated by *AVPR1A*. The *AVPR1A* gene is expressed in the hypothalamus, amygdala, and hippocampus (brain areas involved in estrus behavior), and *AVPR1A* plays a major role in the neurobiology of behaviors such as anxiety, aggression, social recognition and pair bonding [209,210]. Behavioral effects of *AVPR1A* deletion may be related to olfactory deficits [211]. Olfactory stimulation in the presence of a boar is an important stimulant of estrous behavior in gilts, and functional variants that negatively impact this response could lead to reduced expression of pubertal estrus. Also reported was *PAPPA*, pappalysin 1 [192], a metalloproteinase that cleaves insulin-like growth factor binding protein 4 to increase the bioavailability of IGF-I at the tissue level [212], and this would be important for ovarian steroidogenesis during gilt development.

Nonneman et al [193,194] reported a number of candidates related to metabolism and reproduction, including *IGF1R*, *TAC1*, *NELL2*, *ADAMTSL3*, *LEP*, *ESRRG*, *NEBL*, *LHX6*, *ACVR1C*, *NHLH2*, and *SV2B*. Both *IGF1R* and *ESRRG* have been associated with age at first estrus in heifers [213,214]. Several of these genes are involved in glutamatergic and GABAergic function and GnRH secretion. Neural EGFL-like 2 is expressed in glutamatergic neurons with expression maximal during the juvenile nadir but reduced at puberty, and disruption of *NELL2* in the hypothalamus of rats with antisense oligonucleotides reduced GnRH and delayed puberty [215]. Both glutamatergic and GABAergic neurons also express *SV2B* [216]. Nescient helix loop helix 2 (*NHLH2*) is a transcription factor involved in neurogenesis and is expressed in the adult hypothalamus and adenohypophysis. Deleting *NHLH2* in mice causes delayed puberty and reduced sexual behavior owing to hypogonadotropism [217,218] resulting from reduced expression of *GnRH1* in the hypothalamus [219]. A member of the tachykinin family, *TAC1* encodes neurokinin A (also known as substance P) and is expressed in the POA, SCN, and MBH as well as the anterior pituitary gland of the pig [220,221]. Puberty is delayed in mice lacking *TAC1* [222,223], which in mice activates kisspeptin neurons causing secretion of LH; an effect that requires estradiol [224]. It is likely that *TAC1* acts within the porcine hypothalamus in a similar fashion. *TAC1* also directly affects porcine gonadotrope cells to stimulate LH release and potentiates the GnRH stimulation of LH secretion from the anterior pituitary gland [225,226].

Some of the genes reported by Nonneman et al [193,194] regulate secretion of gonadotropin hormones by affecting the function of gonadotrope cells within the adenohypophysis. Serum concentrations of FSH in boars were previously associated with *NEBL* [227], a protein that interacts with the actin cytoskeleton. Gonadotrope cells undergo major remodeling of actin in response to GnRH to secrete gonadotropins into capillaries within the anterior pituitary gland [228,229]. LIM homeobox 6 is a homeodomain protein, and homeodomain response elements contribute to transcriptional activation of the GnRH receptor gene [230]. Activin A receptor type 1C is involved in activin signaling through SMAD pathways, which is essential for transcription of the GnRH receptor gene in pituitary gonadotrope cells [231,232]. Moreover, *ACVR1C* knockout mice display reduced circulating concentrations of FSH and LH and had delayed onset of puberty followed by abnormal cyclicity [233]. Collectively, these genes are probably contributing to greater sensitivity of the pituitary gland to increasing pulses of GnRH at puberty.

7. Summary

Age at puberty is an early indicator of lifetime productivity, and the timing of puberty in gilts is a critical factor in managing the breeding herd for pork production. During the juvenile period, suppression of LH pulses results from negative feedback of estradiol secreted from small antral follicles on the ovary. As gilts mature, they escape this negative feedback, resulting in greater LH pulse frequency that drives maturation of ovarian follicles. Estradiol at the level of the hypothalamus then acts positively to promote

increased GnRH release for an ovulatory surge of LH. Glutamatergic and GABAergic neurons coordinate these events through the changes in the activity of the GnRH pulse generator. Kisspeptin is essential for puberty in pigs and, along with neurokinin B, plays a critical role in stimulating LH secretion during puberty. Positive energy balance, characterized by nutrient sufficiency, is signaled to the hypothalamus through metabolites and metabolic hormones. Leptin acts on NPY and POMC cells in the ARC, which function as metabolic sensors for activation of GnRH secretion. Before puberty, circulating leptin concentrations are low, minimizing the excitatory signal POMC and increasing the inhibitory signal NPY. Remodeling of neuronal projections from NPY to GnRH neurons inhibits GnRH release for high-amplitude, low-frequency LH pulses that limit maturation of ovarian follicles. The inhibitory signal of NPY to GnRH secretion may be amplified by its inhibition of Kiss neuronal stimulation of GnRH secretion. Increasing leptin during pubertal development inhibits the negative effects of NPY on GnRH and Kiss neurons and upregulates activation of POMC cells, which project axonal fibers rostrally to the POA and AHA. These afferent projections are predicted to impact both Kiss and GnRH expressing cells in the POA and ARC, thus driving increased secretion of GnRH for high-frequency, low-amplitude pulses that promote ovarian follicular growth and maturation. The coordination of these events is under genetic control that can be influenced by environment, nutrition, and management. Many of the genetic regions associated with pubertal traits in gilts harbor genes involved in metabolism, adiposity, and activity of the HPG axis. Cumulatively, this illustrates the complex genetic coordination of biological events culminating in activation of the GnRH pulse generator for puberty in gilts (Fig. 1).

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